Latency-Bound Inference

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Core questions of the thesis.

1 **Is this a problem?**

Do today's frameworks miss their deadlines, and where?

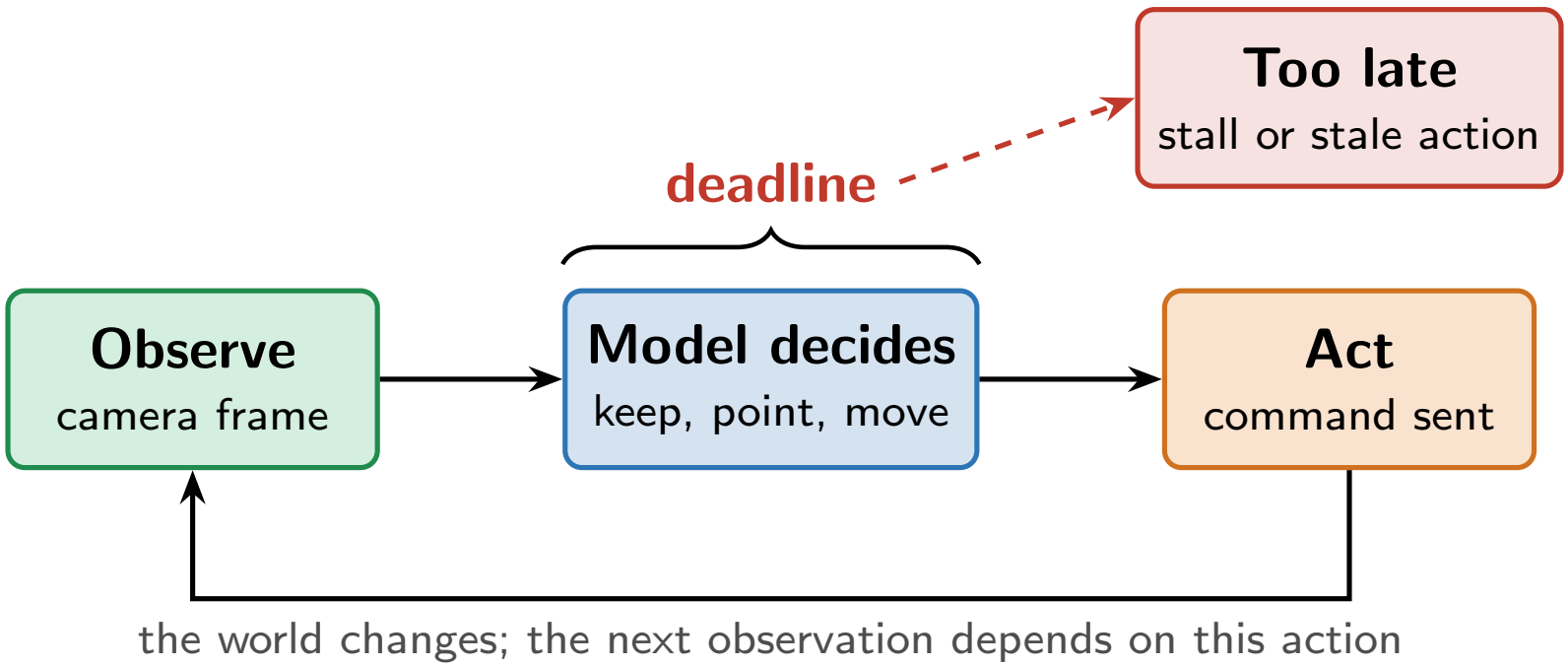
2 **Why is it hard to predict?**

What makes latency vary when the accelerator is shared?

3 **Can software fix it?**

How predictable can GPU inference get, with FPGA timing as the yardstick?

Latency matters in looped settings.



Looped use cases: two in space, two on robots.

We can make inference **faster**, but more importantly **more deterministic**.

Use case	Deadline	Evidence	Slow?	Jitter?
Image triage space, VLM	Before the next image arrives (1–30 s)	26.6 s per image on the ground vs. 20 s between images paper	yes	?
Retasking space, VLM	Before the target leaves view	Kepler says it does this, no numbers yet	?	?
Fast manipulation robot, VLA	About 50 ms	Ping-pong hits: 0/20 sync vs. 11/20 async paper	yes	?
Onboard control robot, VLA	Before the current actions run out	Cloth folding: 0/10 on Orin vs. 7/10 on a 4090 paper	yes	?

Slow is what prior work fixes. **?** means the papers report no jitter, so we do not know if the slow requests are rare or routine.

Prior work: fast on average, or predictable for DNNs only.

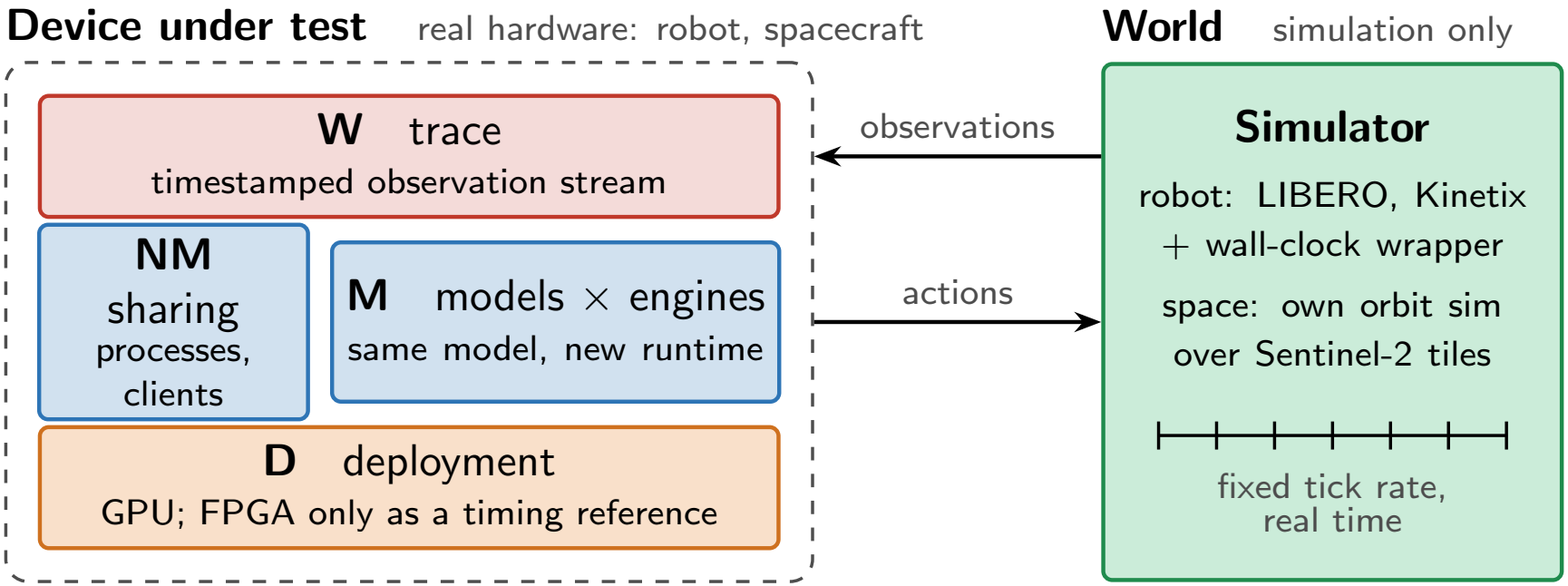
Datacenter LLM serving	Real-time GPU work	Our setting
Fast on average.	**Predictable, but only for DNNs.**	**An LLM in a loop on a Jetson.**
DistServe, Sarathi-Serve, SOLA	Clockwork, REEF, TimeGraph	One late answer breaks the loop

The gap

Nobody measures how often an LLM misses its deadline in a loop on a Jetson.

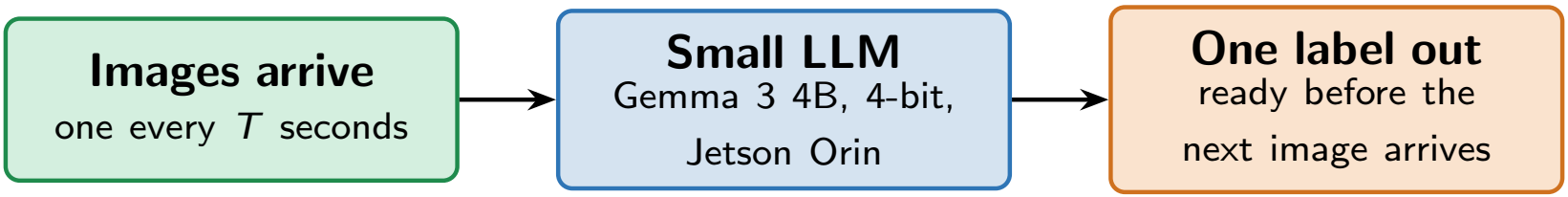
Full table with numbers: appendix.

Measurement setup: real device under test, simulated world.



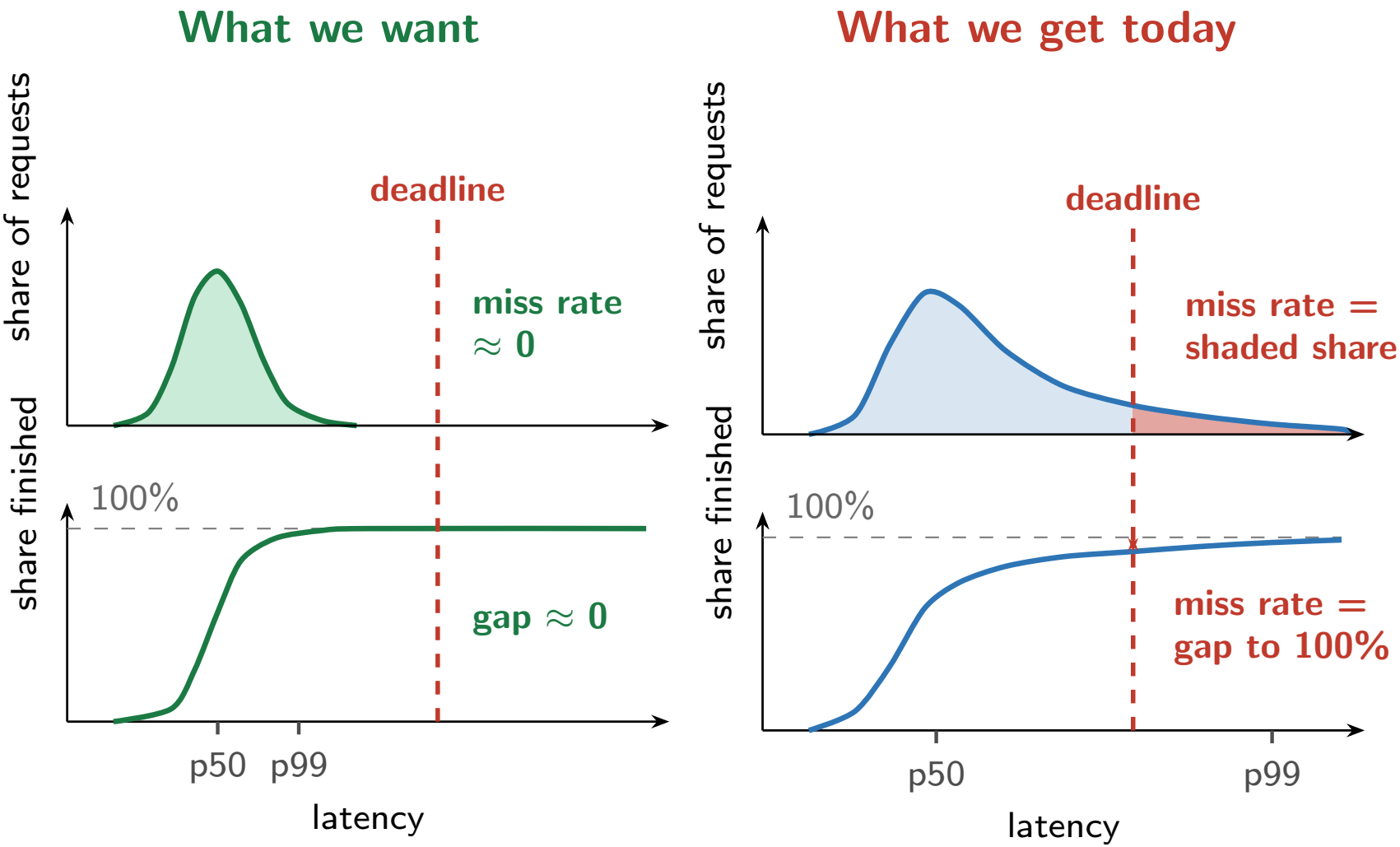
- **Real device, simulated world:** no real robot or orbit in this study.
- **Simulator:** takes actions, steps its world, emits observations.
- **The world never waits for M:** a miss hurts the task, as in reality.

First experiment: a small LLM on a Jetson, fed at a fixed rate.



We vary	We record	We ask
Model alone. Then with other processes. Then at a lower power mode.	Latency of every single request, plus energy and clock speed.	How often is the label late, and which change makes it late?

What we want vs. what we get: the long tail is the miss.



- **Top:** share of requests at each latency. **Bottom:** share finished by then.
- **Miss rate:** the shaded tail on top, the gap to 100% below.

What we measure per request.

Metric	In plain words
Time to first token	How long until the model starts answering. Split into waiting and prefill.
Time between tokens	How steadily the answer streams out.
Time to last token	Total time for the request.
Miss rate	Share of requests that finish after the deadline.
Tail ratio (p99 / p50)	How much worse the slow requests are than the typical one.
Energy per request	Joules, with clock speed, so throttling shows up.

With one-token answers, first token and last token are the same, so only prefill and the miss rate are left.

Appendix

Broader use cases, agentic framing, sources.

Latency-bound inference: use cases from well-known groups.

Domain	Use case	Evidence	Setup
Space	Image filtering	Earth+: only 2% of captured image data can be downloaded (citing Kodan) paper	Constellation
Space	VLM queries	Gemma 3 4B classifies live imagery onboard, April 2026 paper	Jetson Orin AGX
Physical AI	Robot manipulation	VLASH: reaction latency 546–565 ms sync vs. 46–65 ms async paper	RTX 5090, SO-101
Physical AI	Onboard VLA control	Jetson-PI: 8.66× control rate over naive PyTorch paper	Jetson Orin
Physics	Level-1 trigger	hls4ml jet tagger: about 100 ns (2018) paper	FPGA

Agentic use cases: the orbit sets the deadline, not the token rate.

Use case	Agent decision	Deadline (source)	Evidence
Capture triage	Label, keep or drop, one-line description per frame	Inter-capture cadence: 1–30 s (Kodan), 20 s claimed in flight	NAVI-Orbital: Gemma 3 4B, 4-bit, Orin AGX; 26.6 s ground cycle vs. 20 s cadence; retries unbounded paper
Closed-loop retasking	Slew or zoom for the next capture after a wide-FOV detection	Remaining access window: seconds to minutes	Kepler retasking claim (operator, no numbers); the one case where a late decision changes the next observation
Pre-contact packaging	Rank and summarize what to downlink	Contact start, known ahead from SGP4	Earth+: 2% of image data can come down paper; wildfire alerts delayed up to 90 min news
In-pass operator query	Answer an uplinked question over the onboard log	Same contact window (minutes), else next pass	NAVI dialogue agent; output length varies, so tokens trade directly against the deadline

- **Metric:** deadline-miss rate at a fixed budget, plus task quality at that budget. No paper in the review reports a miss rate.
- **Single-token output:** fixed image tokens plus one label is fixed-shape prefill, the regime Jetson-PI's CUDA-graph reuse exploits.

Agentic physical AI candidates from the lit review.

Use case	Why it is agentic	Latency evidence
Navigation among people	Reasoning step picks the route while people move	TIC-VLA / DynaNav: reasoning delay injected paper
Split planner and policy	Large model on a workstation plans, small model on the robot acts	AsyncVLA: enforced latency up to 6 s paper
Subtask sequencing	Pick, fold, place could sit under a planner	Jetson-PI uses one VLA for all three paper

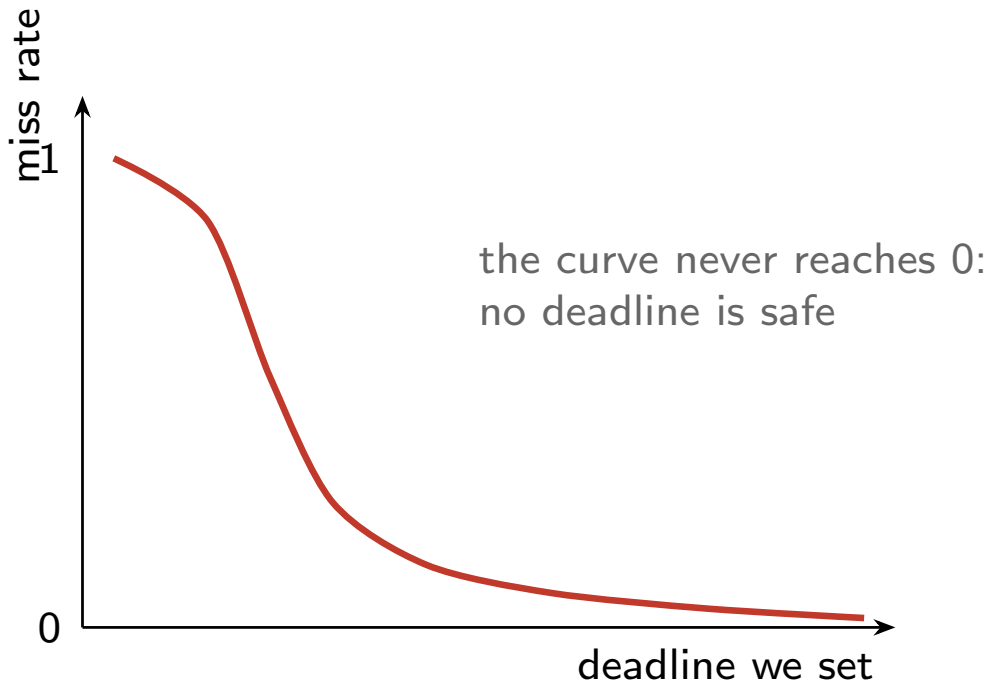
All delays here are injected or enforced. None is a measured deadline-miss rate.

Prior work on meeting deadlines: what exists, and what is missing.

Work	Setting	What it shows	What is missing
DistServe, OSDI '24 paper	Datacenter LLM serving	Splitting prefill and decode serves 7.4× more requests within the SLO	Targets 90% of requests, not all
Sarathi-Serve, OSDI '24, Microsoft paper	Datacenter LLM serving	Chunked prefill stops new requests stalling running ones	Throughput goal, no deadline
SOLA, MLSys '25 paper	Datacenter LLM serving	SLO attainment from 45.5% to 99.4%	Best effort, no guarantee
Clockwork, OSDI '20 paper	DNN serving	100 ms met for 99.997% of requests	Fixed-size DNNs, no decode
TimeGraph, ATC '11, CMU paper	Real-time GPU scheduling	Priorities and reservations protect key GPU tasks	Graphics era, not ML
REEF, OSDI '22 paper	Real-time DNN serving	Preempts GPU kernels in microseconds	DNNs, datacenter GPU

Gap: LLM serving targets averages in datacenters. Real-time GPU work stops at DNNs. Nobody measures LLM misses in a loop on a Jetson.

What deadline can the device promise?



Each point is the miss rate at one deadline. Read it right to left: a longer deadline lowers the miss rate, but a long tail keeps it above zero.

Sources

- Earth+ (ASPLOS 2025)
- NAVI-Orbital: Gemma 3 on Loft YAM-9 (arXiv 2606.18271)
- Loft YAM-9 coverage (news, IEEE Spectrum)
- VLASH, asynchronous VLA inference (arXiv 2512.01031)
- Jetson-PI, onboard VLA control (CoRL 2026)
- hls4ml, fast DNN inference in FPGAs (arXiv 1804.06913)
- TIC-VLA / DynaNav (arXiv 2602.02459, ICML 2026)
- AsyncVLA (Levine group, UC Berkeley; arXiv 2602.13476)
- DistServe (OSDI 2024)
- Sarathi-Serve (OSDI 2024, Microsoft Research)
- SOLA (MLSys 2025)
- Clockwork (OSDI 2020)
- TimeGraph (USENIX ATC 2011); REEF (OSDI 2022)